AI Image Generator

9. Admin Panel

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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, Qwen-Image, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and an Admin panel that manages models, LoRAs and downloads and shows live system statistics.

9. Admin Panel

The Admin tab has two jobs: it manages the studio (models, LoRAs, downloads, settings) and it watches the machine. Everything an admin changes is stored in models/registry.json; everything the panel measures is sampled on demand and never stored. The eight sub-tabs:

Sub-tab	Purpose
Overview	System tiles (9.4), activity tiles (images today and this week, GPU minutes in the last 24 h, loaded and start-up model, models on disk, LoRAs, downloads, outputs size, Hugging Face login) and the per-model statistics table (images, average, fastest and slowest seconds, steps per second, average and maximum peak VRAM, hottest temperature, upscaled count, last use). Refreshes on open, on Refresh and every 5 s when ticked.
Models	Table of every model: key, name, family, type (built-in, local file, added), default steps and guidance, sampler, on disk, size, images, average seconds, enabled, start-up star. Selecting a row fills the editor (9.1).
LoRAs	Table of every LoRA file: name, file, family, trigger words, weight, size, enabled, recorded uses, source. Selecting a row fills the editor (9.2).
Add	Add a model or LoRA from a source (9.3).
Downloads	Every background download with state, bytes, speed, remaining time and the last message; Cancel by number; Clear finished. Refreshes every 3 s.
System	The nine live tiles and six detail sections (9.4 and 9.5).
Settings	Start-up model; default weight for LoRAs ticked in the LoRAs box; default negative prompt; VRAM release after each image on the big flow models; Hugging Face token (saved through huggingface_hub, with Check login); Civitai token; clean-up of generations and images older than N days; admin password lock.
Logs	The app's own log: last N lines with a contains filter, an errors-only level and 3 s auto-refresh.

9.1 Model editor and actions

Field or button	Effect
Name, Description	Shown in the Model box and the model cards.
Family, Source	Editable for added models only; built-ins and local files keep theirs.
Default steps, Default guidance, Sampler, Weight variant	The model's defaults; quality and fidelity words in prompts scale relative to them. Sampler choices: DPM++ 2M Karras, DPM++ SDE Karras, Euler a, Euler, model default.
Aliases, Auto-pick for styles	Extra words that name the model in a prompt; style names for which Auto chooses it when an SDXL-family model is loaded.
Supports a negative prompt, Enabled	Whether the negative box applies; whether the model is listed and usable.
Save changes	Writes the entry (overrides for a built-in, the full spec for an added model). Applies to the next load.
Load now	Loads the model immediately under the GPU lock, downloading first if needed.
Download in background	Queues a download job without loading.
Set as start-up model	The model loaded when the app starts.
Unload current model	Frees the VRAM; the next generation loads again.
Remove from list / reset built-in	Deletes an added model's entry (and unloads it if loaded) or resets a built-in to its defaults. Files stay.
Delete files from disk	With the confirmation ticked: removes the local file or the repo's Hugging Face cache folder and the converted caches under models/cache/.

9.2 LoRA editor and actions

Field or button	Effect
Display name	Shown in the LoRAs box; the prompt tag still uses the file name.
Trained for (family)	Auto switches to a model of this family when the LoRA is requested; on a model of another family the LoRA is skipped with a note rather than loaded.
Trigger words	Shown in the LoRAs box and reminded in the "Will generate" line when missing from the prompt. Civitai sources fill them automatically.
Default weight	Used when the LoRA is ticked in the LoRAs box; a prompt tag such as <lora:name:1.2> overrides it.
Description, Source, Tags, Enabled	Notes; where it came from; free tags; whether it is listed.
Try it in the Studio	Fills the prompt with the trigger words, the LoRA tag, a matching model and a fixed seed, and switches to the Studio tab.
Delete LoRA file	With the confirmation ticked: deletes the file and its metadata; if the adapter is inside the loaded pipeline the model is unloaded so the next load is clean.

9.3 Sources the Add tab understands

Source	Example	What happens
Hugging Face repo, diffusers layout	RunDiffusion/Juggernaut-XL-v9	model_index.json names the pipeline class; StableDiffusionXLPipeline, ZImagePipeline, QwenImagePipeline and FluxPipeline map to the four families. The fp16 variant is used for SDXL when present. Added as a model; download of the needed weights only (about 6.9 of 13.1 GB for Juggernaut).
Hugging Face repo of single files	ostris/crayon_style_lora_sdxl	The largest .safetensors file is selected (others selectable); under 1.5 GB it is treated as a LoRA, above as a checkpoint.
Hugging Face file	nerijs/pixel-art-xl/pixel-art-xl.safetensors or its https link	That file; the stored Hugging Face token is sent for gated repos.
Civitai	https://civitai.com/models/133005?modelVersionId=782002, a model-versions link or an api/download link	The Civitai API supplies type (Checkpoint or LoRA), base model (SDXL 1.0, Pony, Illustrious, FLUX.1 ...), trigger words, file name and size and the NSFW flag; the Civitai token from Settings is used when present.
Direct URL	https://example.com/model.safetensors	HEAD request for name and size; type guessed from the size.
Local path	D:\downloads\model.safetensors	Copied into models/.
Upload	a file chosen in the browser	Moved into models/ (suited to LoRAs; the limit is 8 GB).

Single files are always loaded with the SDXL single-file loader, so FLUX, Qwen-Image and Z-Image additions must be diffusers repos. File downloads resume after an interruption (the .part file is kept) and refuse answers that are web pages, which is what a missing token looks like. Registration happens when the file has arrived, so an interrupted download leaves nothing half-registered.

9.4 System tiles

Tile	Value	Meter or state	Source
GPU load	utilisation %, SM clock, P-state	meter	nvidia-smi
VRAM used	GiB used of total; torch allocation	meter of used %	nvidia-smi, torch
GPU temperature	°C and fan %	state: OK under 70, Warm under 83, Hot under 90, Critical above	nvidia-smi
GPU power	W of limit, % of limit	meter	nvidia-smi

Tile	Value	Meter or state	Source
CPU	usage %, threads, frequency	per-core bar strip	psutil
Memory	GB used of total	meter	psutil
System disk	GB free, % used	meter	psutil
Network	download and upload rate, established connections	-	psutil, two samples
App uptime	uptime, PID, resident memory	-	psutil

Meters use the pink accent and switch to the amber warning colour above 85 % and the red critical colour above 95 %. The temperature state always carries an icon and a word, never a colour alone.

9.5 System sections

Section	Fields
GPU	Card, VRAM, compute capability and SM count; driver, CUDA, cuDNN; current and maximum SM and memory clocks; GPU and memory-bus utilisation; VRAM from nvidia-smi and from torch (allocated, reserved, session peak); PCIe generation and width; throttle reasons decoded from the nvidia-smi bitmask (idle, power cap, thermal slowdown and so on); VBIOS and PCI bus; loaded model and sampler; processes on the GPU.
CPU	Processor name, architecture, physical and logical cores, current and maximum frequency, total and per-core usage, 1/5/15-minute load averages, process and thread counts, boot time and uptime.
Memory and storage	RAM used, available and total; swap or page file; disk reads and writes since boot; outputs folder size and PNG count; models folder; Hugging Face cache size and path; database size; every partition with file system, total, used, free and percentage.
Network	Hostname and primary IP; current throughput (rates need two samples); bytes and packets since boot; errors and drops; established connections and listening ports; every interface with state, IPv4, MAC, link speed, sent and received; optional reachability of huggingface.co, github.com and pypi.org in milliseconds.
Power	GPU draw against the configured limit with percentage; default and maximum limits; performance state and fan; the active Windows power plan; battery if present; estimated GPU energy of all recorded generations in Wh.
Software and app	OS, Python, torch, CUDA, cuDNN, diffusers, transformers, gradio versions; whether the process runs in Docker; app PID, start time, uptime, resident memory and threads; working directory; generation totals from the database.

9.6 Reading the numbers

- Idle readings on the RTX 3090 are about 45 W, P8, SM clock 210 MHz, PCIe gen 1 x1 and throttle reason "GPU idle". Under load expect P0, 1.7 to 1.9 GHz, PCIe gen 3 or 4 x16 and 350 to 390 W.
- nvidia-smi reports allocator reservation as used VRAM; the torch allocated figure is the live tensor total.
- Process names on the GPU may show "Insufficient Permissions" for processes of other users or the system.
- The network rate tiles show 0 on the first sample after start; every later refresh shows the rate since the previous one.
- Inside Docker the CPU, memory and network sections describe the container, while GPU and power come from the host driver.
- A System or Overview refresh costs about 0.7 s (3 s the first time, when folder sizes are computed) and runs on its own queue, so it keeps working while an image is generating.

9.7 Locking the panel

Deleting model or LoRA files and cleaning outputs always need their confirmation box. When the environment variable FRAZILE_ADMIN_PASSWORD is set, those actions also need the panel unlocked once per browser session with that password in Settings. Set it whenever the server is reachable from other machines (Docker, or

GRADIO_SERVER_NAME=0.0.0.0); on localhost it is optional.